

Confluence and Normalization Choice in Exact Pushdown Lifts

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Abstract

A deterministic pushdown computation can have a unique semantic endpoint while still admitting several non-equivalent physical normalization histories. We formulate this distinction as a hierarchy of rewrite problems for exact sectioned pushdown lifts. At the presentation level, enabling one additional admissible eager-normalization site gives a finite terminating rewrite system. For a downward-closed schedule family, this rewrite is confluent exactly when the family has one maximal schedule; for an independence complex, equivalently, every minimal fatal support is a singleton. Applied to the canonical rational $p : q$ schedule families, this yields an exact arbitrary-coprime dichotomy: schedule activation is confluent when $\min(p, q) \geq 3$, or when $\min(p, q) = 1$ and $\max(p, q)$ is odd, and is nonconfluent in the complementary regimes. In every ratio there are exactly one or two schedule normal forms.

We then pass from presentations to physical semantic fibers. Every completion word of residual r has Parikh vector $\mu(r) + k(q, p)$, where $\mu(r)$ is the unique shortest completion demand and k is a primitive-kernel excess. Boundary-oriented adjacent swaps together with deletion of one primitive kernel block form a strongly normalizing confluent rewrite on the entire completion fiber, not only on shortest completions. Its unique normal form is the canonical shortest completion. The proof uses the exact debt rank (k, Ω) , separating quantity excess from word-order debt. An independent finite-control sheet coordinate gives an exact product theorem: termination and confluence of the full physical rewrite reduce precisely to the sheet rewrite, and a rooted sheet rank yields a three-channel debt (k, Ω, η) with one canonical physical normal form.

Finally, we study confluence modulo protected physical information. Bounded bi-recoding is treated as an equivalence, while strict protected coordinate quotients are reductions. A canonical lexicographic anchor in the finite protected Pareto envelope makes protected-core descent strongly normalizing; hence local confluence, global confluence, and uniqueness of the terminal protected recoding class are equivalent. The results separate canonical endpoint coherence, presentation confluence, physical fiber confluence, and protected confluence. Classical rewriting results are used as tools; the contribution is their exact specialization to pushdown normalization and the resulting rational and protected-interface classifications.

Keywords. deterministic pushdown automata; confluence; normalization; Church–Rosser rewriting; rational balance languages; exact lifts; shortest completion; protected interfaces.

1 Introduction

Let

$$L_{p,q} = \{w \in \{0, 1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p, q) = 1.$$

A completion-aware deterministic pushdown realization of this language naturally distinguishes two levels. The semantic level records the weighted residual of the input; the physical level distributes that residual among finite control, stack content, word order, and optional normalization choices. The semantic endpoint can therefore be unique even when the physical chronology is not.

This distinction is easy to blur. If a normalizer is a retraction $N = \sigma\Theta$ from reachable physical configurations to a chosen section of the semantic quotient, then identities such as

$$N^2 = N, \quad N\delta_w = N\delta_w N$$

express canonical endpoint coherence. They do not define a branching rewrite relation, and hence do not by themselves prove a Church–Rosser theorem. Conversely, if every internal normalization step preserves the semantic projection Θ , then “confluence modulo equality of Θ ” is automatic and carries no information about the physical representation.

The goal of this paper is to make the missing rewrite layer explicit. We study three successively stronger settings.

1. *Schedule-presentation rewriting.* Optional eager-normalization sites are enabled one at a time while the resulting schedule remains admissible.
2. *Physical semantic-fiber rewriting.* Local word and finite-control debt is discharged at fixed semantic residual.
3. *Protected rewriting modulo harmless recoding.* Physical information declared by an interface is preserved, while bounded invertible recodings are treated as equivalences rather than reductions.

These levels need not have the same confluence behavior.

The rewrite-theoretic notions used below are classical. Termination, local confluence, confluence, complete rewriting systems, unique normal forms, and Newman’s lemma belong to standard rewriting theory; see, for example, [1, 3]. Oriented commutations such as $ba \rightarrow ab$ are the standard finite complete systems for free commutative structures [4]. Confluence modulo equivalence and invariants has an established literature as well [5]. Our claims are therefore not claims to new general rewrite theory.

There are also two terminology boundaries worth stating at the outset. First, the established class of *Church–Rosser languages* is a different formal-language program, connected to length-reducing systems and shrinking two-pushdown automata [7, 8]. We do not study that language class. We use “Church–Rosser” only in the ordinary rewrite-theoretic sense of confluence for normalization relations associated with one-stack exact pushdown lifts. Second, confluent string rewriting has appeared inside deterministic pushdown analyses: Carton and Perifel use a Noetherian confluent deletion relation to characterize the stack contents of a deterministic pushdown transducer [6]. The present problem is different: the rewrite relation acts on exact normalization fibers and is tied to semantic residuals, shortest completions, schedule geometry, and protected quotient structure.

Main results. At the presentation level we prove a general hypergraph criterion for confluence and apply it to the canonical rational schedule families. At the physical level we obtain a finite convergent rewrite system on every rational completion fiber, with a two-component debt measuring primitive-kernel quantity excess and boundary-order inversions. We then add independent finite-control debt and prove an exact product reduction. Finally, a canonical Pareto anchor repairs a subtle well-foundedness issue in protected-core descent and yields a genuine Newman criterion on protected recoding classes.

From the original 2:1 machine to the confluence problem. The present rewrite problem did not arise from an abstract Church–Rosser question imposed on a pushdown system from outside. Its starting point is the original completion-aware 2:1 DPDA shown in fig. 1. In that machine, finite control and the literal stack jointly carry the remaining completion demand: the side states hold a bounded residual offset, while end-of-input rules materialize that offset into an explicit completion word. The figure is therefore included as the historical and operational origin of the normalization program.

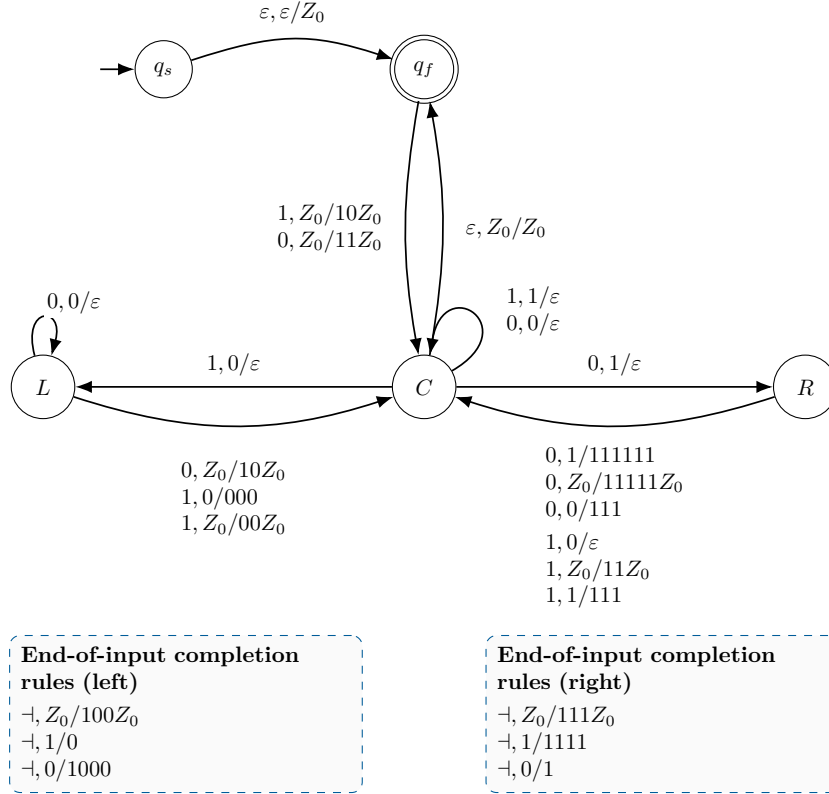


Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included formal variant. Solid arcs process binary input; dashed callout boxes list only end-of-input completion rules. This machine is the provenance and motivation for the normalization-choice problem studied here. The confluence theorems below do not use this transition table as a hypothesis.

The later coprime $p:q$ theory replaces the accidental constants of the 2:1 table by the residual

$$\rho(w) = qN_1(w) - pN_0(w)$$

and the primitive kernel $K = (q, p)$, and selects the unique shortest-completion demand as a canonical section. The WRITE–SEARCH–CANCEL viewpoint then separates the unique semantic carry chronology from the nonunique physical materialization chronology. Eager and lazy realizations of the same semantic update consequently generate a finite schedule space; its causal/fatal geometry provides the presentation-level input to the present confluence problem.

Thus the conceptual route is

$$\begin{array}{c} \text{original 2:1} \longrightarrow p:q \text{ residual/shortest completion} \longrightarrow \text{WSC physical chronology} \\ \longrightarrow \text{causal schedule geometry} \longrightarrow \text{normalization confluence} \end{array}$$

The present paper starts at the final arrow but retains Figure 1 to make the origin explicit. Its arbitrary-coprime and exact-lift theorems are formulated in the generalized objects developed from that origin, not in the historical transition table itself.

2 Exact lifts and four notions of normalization coherence

Definition 2.1 (Exact sectioned lift). *Let M be a deterministic semantic transition system over input alphabet Σ , with semantic transitions $U_a : M \rightarrow M$. An exact sectioned pushdown lift is a reachable physical transition system*

$$\mathcal{L} = (C^{\text{reach}}, \delta, \Theta, \sigma)$$

with semantic projection $\Theta : C^{\text{reach}} \rightarrow M$, physical input transitions δ_a , and section $\sigma : M \rightarrow C^{\text{reach}}$ satisfying

$$\Theta\sigma = \text{id}_M, \quad \Theta\delta_a = U_a\Theta.$$

The canonical retraction is $N = \sigma\Theta$.

Proposition 2.2 (Canonical retraction and absorption; baseline). *The canonical retraction satisfies*

$$N^2 = N, \quad \Theta N = \Theta,$$

and for every input word w ,

$$N\delta_w = N\delta_w N = \sigma U_w \Theta.$$

Thus the canonical endpoint after reading w depends only on the initial semantic state. This is endpoint coherence, not a Church–Rosser theorem for an internal normalization rewrite.

Proof. Since $\Theta\sigma = \text{id}_M$, one has $N^2 = \sigma\Theta\sigma\Theta = N$ and $\Theta N = \Theta$. Exactness gives $\Theta\delta_w = U_w\Theta$, hence

$$N\delta_w = \sigma U_w \Theta = \sigma U_w \Theta \sigma \Theta = N\delta_w N.$$

□

The following distinctions will be used throughout.

Definition 2.3 (Coherence notions). *Let $\Rightarrow$ be an internal normalization relation, distinct from the ordinary input transition δ .*

1. Endpoint coherence: *the designated normalizer sends every reachable configuration to the chosen section representative of its semantic value.*
2. Physical confluence: *every fork under $\Rightarrow$ joins to one physical configuration.*
3. Confluence modulo $\approx$: *every fork joins to configurations equivalent under a declared relation $\approx$.*
4. Section-complete normalization: *every configuration in a semantic fiber reduces to the chosen section point.*

Proposition 2.4 (Bare semantic modulo-confluence is vacuous). *Suppose*

$$c \Rightarrow d \implies \Theta(c) = \Theta(d).$$

Then every fork $c \Rightarrow^ u$, $c \Rightarrow^* v$ already satisfies $\Theta(u) = \Theta(v)$. Thus confluence modulo the equivalence $u \equiv_{\Theta} v$ iff $\Theta(u) = \Theta(v)$ is automatic.*

Proof. Semantic preservation is inherited by the reflexive-transitive closure. Both descendants have semantic value $\Theta(c)$. □

Consequently, a substantive modulo-confluence theorem must protect more physical information than Θ alone.

Definition 2.5 (Strictly canonicalizing physical rewrite). *A tuple $(C, \Rightarrow, \Theta, \sigma, D)$ is strictly canonicalizing if D takes values in a well-founded ordered set and:*

1. $c \Rightarrow d$ implies $\Theta(c) = \Theta(d)$;
2. every rewrite strictly decreases D ;
3. $D(c) = 0$ iff $c = \sigma(\Theta(c))$;
4. every noncanonical c admits a rewrite step.

Proposition 2.6 (Canonical-normal-form criterion). *Every strictly canonicalizing physical rewrite is strongly normalizing and confluent. The unique normal form in the fiber $\Theta^{-1}(m)$ is $\sigma(m)$.*

Proof. Well-founded strict descent forbids infinite chains. Every maximal chain terminates, and the last two conditions imply that a terminal point is exactly the section point in its semantic fiber. Semantic preservation keeps every descendant of a fork in the same fiber, so both branches terminate at the same section representative. □

3 Finite schedule-activation rewriting

Let E be a finite set of optional eager-normalization sites. A schedule is a subset $H \subseteq E$, and $\text{Adm}(E) \subseteq 2^E$ denotes the schedules preserving the intended recognition/normalization contract.

Definition 3.1 (Schedule activation). *Define*

$$H \rightsquigarrow_E H \cup \{e\}$$

when $e \notin H$ and $H \cup \{e\} \in \text{Adm}(E)$. A schedule is normal when no further one-site activation is admissible.

Proposition 3.2 (Termination). *Schedule activation terminates for every finite E .*

Proof. The rank $|E| - |H|$ decreases by one at every step. □

Theorem 3.3 (Exact criterion for downward-closed schedule families). *Assume $\text{Adm}(E)$ is nonempty and downward closed. The following are equivalent:*

1. $(\text{Adm}(E), \rightsquigarrow_E)$ is confluent;
2. there is a unique normal form reachable from $\emptyset$;
3. $\text{Adm}(E)$ has a unique inclusion-maximal member.

Proof. In a downward-closed family, if $H \subseteq K$ are admissible then every intermediate set obtained by adding elements of $K \setminus H$ remains admissible. Hence rewrite normal forms are exactly the inclusion-maximal schedules and every admissible schedule reaches every maximal extension containing it. A unique maximal member therefore joins every branch. Two distinct maximal members are distinct terminal descendants of $\emptyset$ and cannot join. □

Suppose now that admissibility is the independence complex of a finite hypergraph $\mathcal{F}$ of minimal fatal supports:

$$\text{Adm}(E) = \text{Ind}(\mathcal{F}) = \{H \subseteq E : \text{no } F \in \mathcal{F} \text{ is contained in } H\}.$$

Theorem 3.4 (Fatal-support criterion). *Schedule activation is confluent iff every minimal fatal support is a singleton. If*

$$S_1 = \{e \in E : \{e\} \in \mathcal{F}\},$$

then, in the confluent case, the unique normal form is $E \setminus S_1$.

Proof. If every fatal support is singleton, admissibility is exactly $2^{E \setminus S_1}$. Conversely, let $F \in \mathcal{F}$ contain distinct x, y . Minimality makes $F \setminus \{x\}$ and $F \setminus \{y\}$ admissible. Extend each to a maximal admissible schedule. They cannot have the same maximal extension, since such an extension would contain their union F . Thus there are at least two normal forms, and theorem 3.3 gives nonconfluence. □

4 Rational schedule confluence

The canonical baseline-portable literal-demand presentation of rational $p:q$ normalization has a finite optional-site family whose minimal fatal supports are summarized in table 1. Appendix A records the schedule data and the low-width certificates needed for the present rewrite application. The important point here is not the site coordinates themselves but the size of the minimal supports.

Regime	minimal fatal-support structure	maximal-schedule count
$\min(p, q) \geq 3$	all present fatal extras are singleton; the canonical core is free	1
$\min(p, q) = 2$	two non-singleton fork supports $\{A, D\}, \{B, D\}$, plus singleton extras	2
$(p, q) = (1, 1)$	no optional site	1
$\min(p, q) = 1$, odd	all positive optional sites singleton-fatal; one	1
maximum ≥ 3	negative-boundary site is free	
$(2, 1)$ up to duality	$\{B, X\}, \{B, G\}$	2
$\min(p, q) = 1$, even	singleton fatal sites plus the four pairs	2
maximum ≥ 4	between two negative- and two positive-sector sites	

Table 1: Fatal-support geometry needed for the rational schedule-confluence theorem. Physical duality exchanges p and q .

Corollary 4.1 (The canonical 2:1 fork). *For the complete four-site baseline-portable family*

$$E_{2,1} = \{X, A, B, G\},$$

one has

$$\text{Adm}_{2,1} = \{H \subseteq E_{2,1} : \{B, X\} \not\subseteq H, \{B, G\} \not\subseteq H\}.$$

Schedule activation terminates but is not confluent. Its two normal forms are

$$\{A, B\}, \quad \{A, X, G\}.$$

Proof. Appendix A gives the exact minimal fatal supports $\{B, X\}$ and $\{B, G\}$; the site A is free. Hence the maximal admissible schedules are exactly the two displayed sets, and theorem 3.3 gives nonconfluence. $\square$

Theorem 4.2 (Exact arbitrary-coprime schedule-confluence dichotomy). *For every coprime $p, q \geq 1$, activation in the canonical baseline-portable rational schedule family is confluent exactly when*

$$\boxed{\min(p, q) \geq 3 \quad \text{or} \quad \min(p, q) = 1 \text{ and } \max(p, q) \text{ is odd.}}$$

It is nonconfluent exactly when

$$\boxed{\min(p, q) = 2 \quad \text{or} \quad \min(p, q) = 1 \text{ and } \max(p, q) \text{ is even.}}$$

Proof. Read the minimal fatal supports from table 1 and apply theorem 3.4. In the high-minimum and odd boundary regimes all minimal supports are singleton. In the width-two, $(2, 1)$, and even boundary regimes a non-singleton fatal pair is present. $\square$

Corollary 4.3 (One-or-two normal-form law). *Every canonical rational schedule-activation system has exactly one or two normal forms. It has one precisely in the confluent regimes of theorem 4.2 and two precisely in the nonconfluent regimes.*

Proof. The maximal independent sets are visible in table 1. For $\min = 2$, either the fork vertex D is chosen and A, B are excluded, or D is excluded and A, B are chosen, with all free sites added. The $(2, 1)$ and even boundary cases similarly split into exactly two safe sectors. The remaining regimes have one maximal admissible schedule. $\square$

This classification is a presentation-level result. It says that some valid ways of adding eager normalization sites cannot be reconciled by further admissible activations. It does not yet describe rewriting inside a fixed physical semantic fiber.

5 The full rational completion fiber

Fix coprime $p, q \geq 1$ and use the completion equation

$$pN_0(z) - qN_1(z) = r.$$

For each $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r) \in \mathbb{N}_0^2$$

be the unique minimum-length nonnegative solution. The elementary one-dimensional kernel structure gives

$$(x, y) = \mu(r) + k(q, p), \quad k \in \mathbb{N}_0, \quad (1)$$

for every other nonnegative solution (x, y) . Thus $K = (q, p)$ is the primitive redundant completion packet.

Define the canonical shortest completion

$$C(r) = \begin{cases} 1^{y_r} 0^{x_r}, & r \geq 0, \\ 0^{x_r} 1^{y_r}, & r < 0. \end{cases}$$

For $r = 0$ this is ε .

Definition 5.1 (Full completion-fiber rewrite). *Let*

$$\mathcal{C}_{p,q} = \{(r, z) : r \in \mathbb{Z}, z \in \{0, 1\}^*, pN_0(z) - qN_1(z) = r\}.$$

Within a fixed residual fiber, generate $\Rightarrow_{p,q}$ by

$$\begin{aligned} r \geq 0 : & \quad 01 \Rightarrow 10, \quad 1^p 0^q \Rightarrow \varepsilon, \\ r < 0 : & \quad 10 \Rightarrow 01, \quad 0^q 1^p \Rightarrow \varepsilon. \end{aligned}$$

The swap discharges boundary-order debt and the block deletion removes one primitive kernel packet.

Lemma 5.2 (Semantic preservation and exact kernel debt). *Every rewrite in theorem 5.1 preserves r . If $\text{Par}(z) = (x, y)$, then the integer k in (1) is unique and*

$$\mu(r) - \text{Par}(z) = -k(q, p).$$

Thus k is the exact kernel-excess quantity-debt level.

Proof. Swaps preserve Parikh counts. A deletion removes q zeros and p ones, so the value $pN_0 - qN_1$ changes by $pq - qp = 0$. The solution lattice of the coprime equation is generated by (q, p) , and minimality of $\mu(r)$ forces the coefficient to be nonnegative. $\square$

Theorem 5.3 (Full completion-fiber confluence). *For every coprime $p, q \geq 1$ and every residual r , the rewrite of theorem 5.1 is strongly normalizing and confluent on the entire completion fiber. Its unique normal form is $C(r)$.*

Proof. Define the sign-adapted inversion number

$$\Omega_r(z) = \begin{cases} \#\{i < j : z_i = 0, z_j = 1\}, & r \geq 0, \\ \#\{i < j : z_i = 1, z_j = 0\}, & r < 0, \end{cases}$$

and let $k(z)$ be the kernel level from theorem 5.2. Use lexicographic order on

$$D_r(z) = (k(z), \Omega_r(z)) \in \mathbb{N}_0^2.$$

A swap fixes k and decreases Ω_r by one. A kernel deletion decreases k by one. Hence every rewrite strictly decreases D_r , so the system is strongly normalizing.

Let z be irreducible. If $r \geq 0$, absence of 01 forces $z = 1^y 0^x$; if $r < 0$, absence of 10 forces $z = 0^x 1^y$. Write $(x, y) = \mu(r) + k(q, p)$. If $k \geq 1$, then the block junction contains $1^p 0^q$ in the nonnegative orientation, respectively $0^q 1^p$ in the negative orientation, contradicting irreducibility. Therefore $k = 0$ and $z = C(r)$. Conversely $C(r)$ has neither an order redex nor a removable kernel packet. Thus every fiber has one normal form, and termination implies confluence (equivalently, apply theorem 2.6). $\square$

Corollary 5.4 (Shortest-completion stratum). *On the stratum $k = 0$, kernel deletion is unavailable and theorem 5.3 reduces to the boundary-oriented adjacent-swap normalization of shortest completions. The theorem therefore extends geodesic normalization to every nonminimal kernel level.*

Corollary 5.5 (Two-channel physical debt). *The pair*

$$(k(z), \Omega_r(z))$$

is an exact normalization debt on completion words: its first coordinate measures redundant primitive-kernel quantity and its second coordinate measures word-order debt.

The point of theorem 5.3 is physical, not merely semantic. The local relation has many possible reduction orders; the theorem proves that all of them reach the same literal word. This is strictly stronger than applying the retraction $N = \sigma\Theta$ after the fact.

6 Finite-control sheets and three-channel confluence

Completion words have no independent finite-control coordinate. To model the third debt channel, let W_r be the word fiber above and let $(S_r, \rightarrow_r)$ be a separately declared control-sheet rewrite system with distinguished sheet s_r^* .

Definition 6.1 (Sheet-separable normalization). *Set*

$$\widehat{C}_r = W_r \times S_r, \quad \widehat{\sigma}(r) = (C(r), s_r^*).$$

The combined rewrite is generated by

$$(z, s) \Longrightarrow_r (z', s) \quad \text{if } z \Rightarrow_{p,q} z',$$

and

$$(z, s) \Longrightarrow_r (z, s') \quad \text{if } s \rightarrow_r s'.$$

The two move types commute exactly because they act on disjoint coordinates.

Theorem 6.2 (Exact product reduction). *For every residual r :*

1. $\Longrightarrow_r$ terminates iff $\rightarrow_r$ terminates;
2. $\Longrightarrow_r$ is confluent iff $\rightarrow_r$ is confluent;
- 3.

$$\text{NF}(\widehat{C}_r) = \{C(r)\} \times \text{NF}(S_r);$$

4. every point of $\widehat{C}_r$ reduces to $(C(r), s_r^*)$ iff every sheet reduces to s_r^* .

Proof. The word relation is terminating and confluent with one normal form by theorem 5.3. An infinite combined chain would contain infinitely many moves in one coordinate; this is possible exactly when the sheet relation has an infinite chain. For confluence, project a combined fork to the two coordinates. Word projections join by theorem 5.3, sheet projections join exactly when $\rightarrow_r$ is confluent, and exact cross-commutation combines the joins. Conversely, any sheet fork lifts to a fork from $(C(r), s)$, so combined confluence projects back to sheet confluence. Normal forms and section reachability are coordinatewise. $\square$

Corollary 6.3 (Confluence does not imply section-complete normalization). *Let S_r contain at least two sheets and let $\rightarrow_r$ be empty. Then the product rewrite is terminating and confluent but has one normal form $(C(r), s)$ for each sheet.*

Proof. The empty sheet relation is terminating and confluent. Apply theorem 6.2. $\square$

Thus

branching coherence $\not\Rightarrow$ collapse of the whole semantic fiber to the section.

Theorem 6.4 (Rooted-sheet canonicalization). *Suppose there is a well-founded rank $\eta_r : S_r \rightarrow W_r^{\text{sh}}$ such that every sheet move strictly decreases η_r , $\eta_r(s) = 0$ iff $s = s_r^*$, and every noncanonical sheet admits a move. Then the combined rewrite is strictly canonicalizing with lexicographic debt*

$$\widehat{D}_r(z, s) = (k(z), \Omega_r(z), \eta_r(s)).$$

It is therefore strongly normalizing and confluent, with unique normal form $(C(r), s_r^)$.*

Proof. Kernel deletion decreases the first coordinate, sorting decreases the second while fixing the first, and sheet discharge decreases the third while fixing the first two. The debt is zero exactly at the section point. Apply theorem 2.6. $\square$

The separability hypothesis is deliberate. We do not assert that an arbitrary DPDA configuration fiber decomposes as word $\times$ sheet. The theorem identifies a natural class in which the independent control channel can be handled without semantically erasing it.

7 Protected confluence and canonical Pareto anchoring

We finally allow harmless changes of physical code while retaining declared interface information. The minimum structure needed for the argument is stated abstractly so that the protected confluence theorem is self-contained.

Definition 7.1 (Protected exact-lift class). *Fix a class of finite-coordinate exact sectioned lifts of one semantic system. Each lift $\mathcal{L}$ carries a finite protected interface*

$$\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\text{reach}} \rightarrow \Lambda.$$

A bounded bi-recoding (BBR) is an invertible bounded physical recoding of finite control and reachable stack blocks that intertwines semantics, section, input transitions, and λ . We write $[\mathcal{L}]$ for its BBR class.

A strict protected coordinate quotient (SCQ) is a surjective coordinate quotient preserving semantics, transitions, section, the unique bottom marker, and λ , and inducing a componentwise nonincreasing map on the used resource coordinates, with at least one strict decrease:

$$\mathbf{R}(\mathcal{K}) < \mathbf{R}(\mathcal{L}) \quad \text{componentwise, with at least one strict coordinate,}$$

where $\mathbf{R}(\mathcal{L}) = (|Q_{\text{use}}|, |\Gamma_{\text{use}}|)$.

For a BBR class $X = [\mathcal{L}]$, define its protected recoding spectrum

$$\mathcal{R}_{\lambda}(X) = \{\mathbf{R}(\mathcal{L}') : \mathcal{L}' \simeq_{\text{BBR}, \lambda} \mathcal{L}\},$$

and its Pareto envelope

$$\mathcal{P}_{\lambda}(X) = \text{Min } \mathcal{R}_{\lambda}(X) \subset \mathbb{N}^2.$$

Dickson's lemma implies that this antichain is finite. Every member is realized by an actual representative because it is a spectrum point.

Definition 7.2 (Protected modulo-equivalence). *Let X be the state set of a normalization rewrite and let*

$$\tau : X \rightarrow \Lambda^\sharp$$

be a declared protected signature. An equivalence $\approx_\tau$ is protected when

$$x \approx_\tau y \implies \tau(x) = \tau(y).$$

At configuration level one may take $\tau = \lambda$. When the rewrite states are BBR classes, τ may be any BBR-invariant signature induced by the protected interface, such as the isomorphism type of a specified finite interface-labelled observable. Passing from representatives to BBR classes already quotients out BBR equivalence itself.

Proposition 7.3 (Terminal protected fork). *In a terminating rewrite presentation equipped with a protected signature τ , if a source has terminal descendants u, v with $\tau(u) \neq \tau(v)$, then the system is not confluent modulo any protected equivalence $\approx_\tau$.*

Proof. Terminality leaves no further reduction. A modulo-join would therefore force $u \approx_\tau v$, contradicting preservation of τ . $\square$

A subtlety arises if each reduction step is allowed to choose an arbitrary incomparable Pareto point of the current BBR class: one no longer has a single resource pair that visibly decreases along every possible path. We remove this hidden source choice canonically.

Definition 7.4 (Lex-normalized protected-core rewrite). *Let*

$$\nu_\lambda(X) = \min_{\text{lex}} \mathcal{P}_\lambda(X).$$

A step

$$X \implies_\lambda^{\text{lex}} Y$$

chooses a BBR-equivalent representative of X attaining the actual resource pair $\nu_\lambda(X)$, takes a strict protected SCQ quotient to a lift $\mathcal{K}$, and sets $Y = [\mathcal{K}]$.

Theorem 7.5 (Strong normalization of canonical protected-core descent). *Every lex-normalized protected-core step strictly decreases ν_λ in lexicographic order on $\mathbb{N}^2$. Hence $\implies_\lambda^{\text{lex}}$ is strongly normalizing.*

Proof. Put $r = \nu_\lambda(X)$. A strict protected SCQ quotient from a representative attaining r has raw resource vector $r' < r$ componentwise, with at least one strict coordinate. The quotient is a representative of the target BBR class Y . Hence some Pareto point $r^\sharp \in \mathcal{P}_\lambda(Y)$ satisfies $r^\sharp \leq r'$. By lexicographic minimality,

$$\nu_\lambda(Y) \leq_{\text{lex}} r^\sharp \leq_{\text{lex}} r' <_{\text{lex}} r.$$

Lexicographic order on $\mathbb{N}^2$ is well-founded. $\square$

Theorem 7.6 (Protected-core confluence criterion). *On the descendant cone of any starting BBR class under $\implies_\lambda^{\text{lex}}$, the following are equivalent:*

1. *confluence;*
2. *local confluence;*
3. *exactly one terminal BBR class is reachable from the starting class.*

Proof. Termination is theorem 7.5, so Newman's lemma gives equivalence of confluence and local confluence [2, 1]. Every descendant of a terminating system reaches a terminal class. Confluence forces all terminal descendants to coincide. Conversely, if one terminal class T is reachable from the source and is the only terminal descendant of that source, then every descendant also terminates at T , because any terminal descendant of a descendant is a terminal descendant of the source. Thus any two descendants join at T . $\square$

Corollary 7.7 (Protected signature certificate). *If two terminal lex-normalized descendants have different values of a BBR-invariant protected signature τ , then the protected-core rewrite is nonconfluent from that source (and cannot be repaired by passing to any equivalence that preserves τ).*

Proof. Apply theorem 7.3. □

The canonical anchor is essential to the theorem as stated. We do not claim that a larger unanchored relation allowing arbitrary re-selection of incomparable Pareto points is strongly normalizing or confluent.

8 Related work and novelty boundary

The paper uses several classical tools whose status should be separated from the normalization-specific results.

Complete rewriting systems. The basic definitions and the implication from termination plus local confluence to confluence are classical [2, 1, 3]. We do not claim new abstract rewrite theory. Likewise, oriented adjacent swaps and their block-sorted normal forms are standard in complete rewriting systems for free commutative monoids; Hermiller and McCammond classify the finite reduced complete systems in that setting [4]. The algebraic skeleton of theorem 5.1 is therefore not, by itself, a novelty claim.

The DPDA-specific statement in theorem 5.3 is the identification of all nonnegative completion words of a fixed rational residual with the levels $\mu(r) + k(q, p)$, the sign-dependent section dictated by shortest completion, and the exact two-channel debt (k, Ω_r) showing how a convergent physical normalization acts on that semantic fiber.

Confluence modulo equivalence. Confluence modulo an equivalence relation and under invariants is established in rewrite-based programming, including the necessary-and-sufficient criteria developed for Constraint Handling Rules by Gall and Fruehwirth [5]. Our protected result uses that general viewpoint but specializes the equivalence to bounded bi-recoding of exact pushdown lifts and couples it to a protected Pareto envelope and strict coordinate quotients. The lexicographic anchor is introduced to make this particular quotient descent independently well founded.

Church–Rosser languages. Church–Rosser Thue systems and the associated formal-language class have a substantial literature [7, 8]. That terminology concerns a different object; in particular, shrinking two-pushdown automata enter characterizations of the language class. The present paper concerns internal normalization rewrites for one-stack exact lifts and should not be read as a statement about membership in the Church–Rosser language class.

Confluent rewrites in pushdown analyses. Carton and Perifel use a Noetherian confluent deletion relation $aa \rightarrow \varepsilon$ to characterize stack contents in a deterministic pushdown compression construction [6]. This provides a useful adjacent precedent for using a confluent word rewrite to organize pushdown stack behavior. Here the target is instead normalization choice: schedule presentations, exact residual completion fibers, finite-control debt, and protected quotient cores.

A targeted literature search located these adjacent theories but no direct prior theorem matching the complete chain developed here. Accordingly, the novelty claim is deliberately package- and specialization-level rather than a priority claim over general confluence methods.

9 Computational regression

The central proofs are symbolic. A direct finite rewrite-graph program was nevertheless used to test sign, boundary, and irreducibility conventions in theorem 5.3. The regression covered all coprime pairs

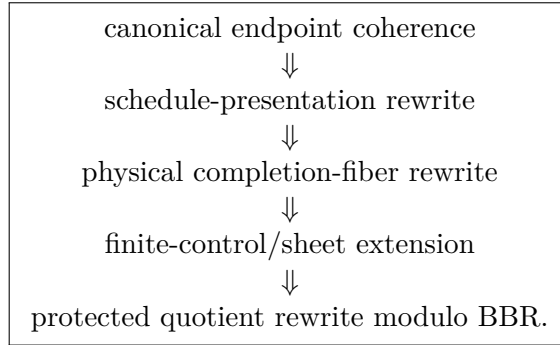
$$1 \leq p, q \leq 4,$$

all residuals $|r| \leq 6$, kernel levels $0 \leq k \leq 2$, and all completion words of total length at most 10. In total, 8,751 completion words were enumerated. Every maximal rewrite branch terminated at the predicted $C(r)$ and no discrepancy was found. The computation is not used as proof.

The presentation-level dichotomy was independently cross-checked against the complete rational fatal-support table: each high-minimum or odd boundary instance has one maximal admissible schedule, and each width-two or even boundary instance has exactly two. Again this is a regression of the exact support classification, not a substitute for it.

10 Synthesis and conclusion

The results give a strict hierarchy of normalization statements:



The arrows indicate increasing physical structure, not logical implication in both directions. Endpoint coherence alone does not control arbitrary normalization branches. Presentation activation can be nonconfluent even though every admissible presentation has the same semantic endpoint. Physical confluence can hold while an inert finite-control sheet leaves several physical normal forms in one semantic fiber. A rooted sheet rank restores section-complete normalization. At the protected level, a canonical resource anchor is needed before Newman’s lemma can be applied without hiding a choice among incomparable Pareto representatives.

For rational normalization, the schedule layer is completely classified for all coprime ratios, while the whole completion-word fiber admits a canonical convergent physical rewrite. These two facts are complementary: one describes nonuniqueness among admissible normalization *presentations*; the other describes uniqueness of normalization *inside* a fixed residual completion fiber. The protected result explains how much of this uniqueness survives when harmless physical recodings are factored out but declared interface data remains visible.

No universal confluence theorem for arbitrary DPDA presentations is asserted. The exact statements are tied to the explicitly typed rewrite relations above. This is precisely the point: confluence in pushdown normalization becomes meaningful only after the object being rewritten, the preserved semantics, and the permitted physical equivalence have all been specified.

A Rational fatal-support data used in Section 4

This appendix makes the rational schedule application self-contained at the level needed by the confluence theorem: it specifies the optional-site contract, the complete site universe, and the

exact minimal-fatal-support data. No theorem in the abstract rewrite sections depends on these coordinates.

Put

$$m = p + q, \quad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \quad K = (q, p),$$

and let

$$\mu(r) = \arg \min \{x + y : x, y \in \mathbb{N}, px - qy = r\}$$

be the unique minimum nonnegative demand vector. Write $\eta(x, y)$ for the fixed canonical literal stack word representing the demand pair (x, y) . The centered literal-demand lift has debt states $D_{-h}, \dots, D_h$ and center state C ; write $V_d = \mu(dm)$. A local work-top observation is a triple

$$o = (d, a, z), \quad |d| \leq h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d , the next input letter is a , and the visible stack symbol is z . With $e_0 = (1, 0)$, $e_1 = (0, 1)$, define

$$B_o = e_z - e_a + V_d, \quad \kappa(o) = \max \left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}, \quad R_o = B_o + \kappa(o)K.$$

The optional eager candidate at o consumes a , replaces the visible cell by the literal minimum-demand word representing R_o , and returns directly to C . This is the unique shortest nonnegative direct return preserving the residual invariant, because every other direct-center replacement belongs to the affine line $B_o + \mathbb{Z}K$.

A *baseline episode* begins at a canonical center demand reached by a mandatory WRITE, then follows only mandatory MATCH/SEARCH transitions, and ends before the next mandatory bottom or outer WRITE. A site is *baseline-portable* when its eager rule lands on the canonical center representative for every source exposed in every baseline episode. A schedule is a subset of the resulting finite menu; it is admissible when the deterministic machine still recognizes exactly $L_{p,q}$. A set F of sites is *minimal fatal* when every schedule containing F is bad but no proper subset has that property. Thus the terms used in table 1 refer to a completely specified finite presentation rather than to unnamed abstract vertices.

A.1 Complete site universe for $\min(p, q) \geq 2$

By physical duality it suffices to take $p \geq q \geq 2$. If m is odd, the complete baseline-portable universe consists of the six core sites

$$(-h, 0, 0), (-h, 1, 0), (-h, 1, 1), (h-1, 0, 0), (h-1, 1, 0), (h, 0, 0),$$

together with

$$\begin{aligned} X = (h-q+1, 0, 1) &\iff p > 5q-5, \\ Y = (h-q, 1, 1) &\iff p > 5q-1. \end{aligned}$$

If m is even, the complete universe consists of the four core sites

$$(-h, 1, 1), (h, 0, 0), (h, 0, 1), (h, 1, 1),$$

together with

$$\begin{aligned} X = (h-q+1, 0, 0) &\iff p > 5q-5, \\ Y = (h-q, 1, 0) &\iff p > 5q. \end{aligned}$$

Hence the menu has at most eight sites. For $q > p$, apply

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1-a, 1-z).$$

For completeness, the exclusion mechanism is exact rather than computational. For the bottom-WRITE seed family

$$B_j = \mu(jm + q)$$

one has, for $j \geq 0$,

$$B_j = (j + q, p - j - 1),$$

while for $j = -t < 0$, writing $t = kq + s$ with $0 \leq s < q$,

$$B_{-t} = \begin{cases} (0, km - 1), & s = 0, \\ (q - s, p + km + s - 1), & 1 \leq s < q. \end{cases}$$

For $1 \leq d \leq h$, $V_d = (d + q, p - d)$ and the three nontrivial positive-debt compiler count vectors have imbalances

$$2d + q - p, \quad 2d + q - p + 2, \quad 2d + q - p - 2.$$

If $m = 2h + 1$, the central seed $B_{h-q} = (h, h)$ has word $(10)^h$. At an interior debt d , the untouched suffix is $(10)^{h-d-1}$ below a visible 0 and $0(10)^{h-d-1}$ below a visible 1. A top-0 eager concatenation can be canonical only when its compiler word is balanced, while a top-1 concatenation requires exactly one excess 1; solving these conditions leaves precisely the two odd-parity candidates displayed above. If $m = 2h + 2$, the central seed is $(h, h + 1)$ with word $(10)^h 1$; the corresponding parity calculation eliminates every top-1 interior candidate and leaves precisely the two displayed top-0 candidates. On the negative side, every nontrivial compiler has a terminal 1-tail whereas the diagnostic exposes an untouched 0, so no negative-interior observation is portable.

The boundary source fibers give exactly the stated cores. For the two odd-parity survivors the common compiler word is $(10)^h 1$ and the extremal exposed one-counts are $2q - 2$ and $2q$, so kernel reduction is equivalent to

$$h + 2q - 2 < p, \quad h + 2q < p,$$

which simplify to $p > 5q - 5$ and $p > 5q - 1$. In even parity the common compiler is $(10)^{h+1}$, the extremal counts are $2q - 3$ and $2q - 1$, and the same calculation gives $p > 5q - 5$ and, using coprimality of odd p, q , $p > 5q$. Thus no unlisted optional site is being silently ignored.

A.2 High-minimum fatal supports

Assume $\min(p, q) \geq 3$. Every subset of the parity-dependent core above is admissible. Indeed, induction over reset episodes keeps every core source inside a protected unary envelope, and every enabled core activation returns to a canonical minimum-demand word. Minority counts stay below the opposite boundary threshold, so a core reset cannot create a new incompatible source on the other side.

Each present extra site X or Y is singleton-fatal. Repeated self-activation increases a hidden alternating prefix until the first nonminimal target is reached. At that first failure the target is exactly

$$\mu(r) + K,$$

not $\mu(r) + jK$ with $j \geq 2$. Feeding a shortest semantic completion for r removes the minimum component and leaves exactly one balanced primitive-kernel packet physically present at semantic residual zero. The witness for Y meets no other optional site; the witness for X meets no core site and at most the neighboring Y observation. Therefore the singleton fatality is robust under every larger schedule. Consequently the only minimal fatal supports in this regime are the present singleton extras.

A.3 Width two

By duality take $q = 2$ and odd $p \geq 3$, with the six odd-parity core sites labelled

$$A = (-h, 0, 0), B = (-h, 1, 0), C = (-h, 1, 1), D = (h-1, 0, 0), E = (h-1, 1, 0), F = (h, 0, 0).$$

The core has exactly the two non-singleton minimal supports

$$\{A, D\}, \quad \{B, D\}.$$

The possible extras are

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \quad X_2 = (h-2, 1, 1) \quad (p \geq 11),$$

and each present extra is singleton-fatal. Hence

$$\text{Adm}_{p,2} = \{H : X_1, X_2 \notin H, \{A, D\} \not\subseteq H, \{B, D\} \not\subseteq H\},$$

with absent extras omitted.

Here the sufficiency and necessity can be checked directly. Write $p = 2h - 1$, $m = 2h + 1$, and for $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \quad k_t = \left\lceil \frac{t}{2} \right\rceil, \quad \beta_t = t + pk_t, \quad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The mandatory reset family is explicitly

$$\mathcal{R}_h^{\text{base}} = \{\mu(jm + 2) : -h - 1 \leq j \leq h\} \cup \{(x, h - 2) : x \geq h + 3\} \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}.$$

Two episode-closed repertoires are

$$\begin{aligned} \mathcal{R}_{-D,h} &= \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h - 1) : x \geq h + 2\}, \\ \mathcal{R}_{D,h} &= \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(x, h - 1) : x \geq h + 2\} \cup \{(x, h) : x \geq h + 1\}. \end{aligned}$$

Every pair in either repertoire has minority coordinate at most h . Before the first reset, MATCH and interior SEARCH only delete the visible top cell, so a boundary source is a suffix of its starting canonical word. This forces the exact source geometry: with D disabled, the common A/B source at D_{-h} is the one-cell stack 0; with D enabled it is an arbitrary 0^n , $n \geq 1$; all C sources are unary 1^n and all D, E, F sources are unary 0^n .

The eager targets are therefore exactly

site	canonical target pair
A	(ϵ_h, β_h)
B	$(\epsilon_{h-1}, \beta_{h-1})$
C	$(\epsilon_h, \beta_h + n - 1)$
D	$(n + h, h)$
E, F	$(n + h + 1, h - 1).$

Thus $\mathcal{R}_{-D,h}$ is closed under every subset of A, B, C, E, F , giving $2^5 = 32$ safe schedules with $D = 0$, while $\mathcal{R}_{D,h}$ is closed when $D = 1$ and $A = B = 0$, giving another $2^3 = 8$. This proves the 40 safe schedules without enumeration.

For necessity, the D -ray gives, for every $x \geq h + 1$,

$$(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the A/B source fiber from $\{0\}$ to the whole unary ray. A uniform prefix reaches $(D_{-h}, 000)$ while meeting no optional site except D . At that common source

$$A : (D_{-h}, 000) \xrightarrow{0} (C, \Gamma_h 00), \quad B : (D_{-h}, 000) \xrightarrow{1} (C, \Gamma_{h-1} 00).$$

Each Γ_t ends in at least p consecutive ones; writing $\Gamma_t = z_t 1^p$, subsequent MATCHes along z_t leave the nonempty stack $1^p 00$. Its residual is $2p - 2p = 0$, so it is one primitive kernel packet at semantic zero and is nonaccepting. Hence $\{A, D\}$ and $\{B, D\}$ are robust minimal fatal supports. The two maximal core schedules are $\{A, B, C, E, F\}$ and $\{C, D, E, F\}$; the extra sites are independently singleton-fatal.

A.4 The boundary line $q = 1$

For $(p, q) = (1, 1)$ the optional-site universe is empty. For $(2, 1)$ the complete four-site universe is

$$X = (0, 1, 0), \quad A = (-1, 1, 1), \quad B = (-1, 0, 0), \quad G = (1, 0, 0),$$

and the minimal fatal supports are

$$\{B, X\}, \quad \{B, G\}.$$

The balanced words 111001 and 111110001 are explicit fatal certificates; they end at semantic residual zero with a nonaccepting ordering of one primitive kernel packet. There are exactly ten admissible schedules and two maximal ones.

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \quad D = (h, 1, 1), \quad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h - 1, 1, 0), \quad C = (h, 0, 1).$$

Every positive optional site is singleton-fatal, while A alone generates an episode-closed unary negative reset corridor. For $h \geq 2$ one may use

$$w_B = 1^{p(p-2)} 0^{p-2}, \quad w_C = 1^{2h(h+1)-1} (0 1^{p-1})^{h-1} 0^h,$$

and for every $h \geq 1$,

$$w_D = 1^{2h(h+1)} 0^p 1^{2h(h+1)+1}, \quad w_E = 1^{2h^2+4h+1} 0^{p+1} 1^{2h(h+1)+1}.$$

The B -witness meets no other optional site; after B is excluded the C - and D -witnesses are unpreempted; after B, D are excluded the E -witness is unpreempted. The last two end at the noncanonical zero-residual kernel word $(C, 1^p 0)$, and the first two likewise end at explicit nonaccepting zero-residual configurations. Hence no positive-site failure can be repaired by enlarging the schedule, whereas the unary corridor generated by A stays reset-closed. Thus

$$\text{Adm}_{p,1} = \{\emptyset, \{A\}\},$$

so there is one maximal schedule.

For even $p = 2h \geq 4$, the always-present sites are

$$A = (-h, 1, 1), \quad B = (-h, 0, 0), \quad C = (-h, 1, 0), \quad E = (h-1, 0, 0), \quad F = (h-1, 1, 0), \quad G = (h, 0, 0),$$

and $D = (h-1, 1, 1)$ is also present when $p \geq 6$. The vertices D and E are singleton-fatal when present, and the non-singleton minimal supports are exactly

$$\{B, F\}, \quad \{B, G\}, \quad \{C, F\}, \quad \{C, G\}.$$

After singleton-fatal sites are removed, every safe schedule lies in one of the two closed sectors

$$\mathcal{N} = \{A, B, C\}, \quad \mathcal{P} = \{A, F, G\}.$$

For subsets of $\mathcal{N}$, the only new optional reset words are unary-one words and the source fibers satisfy $\mathcal{E}_A \subseteq 1^+$ and $\mathcal{E}_B = \mathcal{E}_C = \{0\}$; the mandatory reset family plus this unary ray is episode-closed. For subsets of $\mathcal{P}$, the only new positive reset ray is $(10)^h 0^n$, $n \geq 1$, together with the unary-one negative ray from A ; all positive sources remain unary zero and all targets remain canonical and kernel-reduced. Thus every subset of either sector is safe.

Necessity has uniform balanced certificates. For $h \geq 3$ one may take

$$w_D = 1^{p(p-3)} 0^{p-3},$$

and for $h \geq 2$,

$$w_E = 1^{(h+1)(p-1)} (0 1^{p-1})^{h-1} 0^h.$$

The four cross-pair supports are certified by

$$\begin{aligned} w_{BF} &= 1^{h(p+1)} 0^p 1^{h(p-1)}, & w_{CF} &= 1^{h(p+1)} 0^{p-1} 1^{h(p-3)}, \\ w_{BG} &= 1^{h(p+3)} 0^{p+1} 1^{h(p-1)}, & w_{CG} &= 1^{h(p+3)} 0^p 1^{h(p-3)}. \end{aligned}$$

In each cross-pair certificate the positive site first expands the negative-boundary top-0 source from 0 to 00, after which the indicated negative site traps exactly one primitive kernel packet. Consequently the only maximal schedules are $\mathcal{N}$ and $\mathcal{P}$, and the union of their subset families has $8 + 8 - 2 = 14$ schedules because the overlap is $2^{\{A\}}$. Physical duality gives the cases $p = 1$.

Combining the four regimes proves the exact fatal-support table used in table 1. In particular, its use in theorem 4.2 is no longer an appeal to an unnamed external classification: all vertices, the admissibility contract, the exhaustive universe, and the fatal-support mechanisms required by the confluence argument have been specified here.

Statements and Declarations

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Data and code availability. The paper is theoretical and uses no empirical dataset. All definitions and symbolic arguments required for the results are contained in the manuscript. The finite computations reported in Section 9 are auxiliary regression checks and are not used as proof.

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